Volume Reduction — 3/3

For general case: $E = E(A,c)$, $a \in \mathbb{R}^n$

We can find an invertible affine transformation

T that maps:

$$T[E] = F = E(I,0), \quad \text{and} \quad a \mapsto \begin{pmatrix} -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

For any ellipsoids E_1, E_2, and any invertible affine transformation,

$$\boxed{\frac{\mathrm{vol}(T(E_1))}{\mathrm{vol}(T(E_2))} = \frac{\mathrm{vol}(E_1)}{\mathrm{vol}(E_2)}}$$

— we are back to the special case. ∎

Bounds on initial radius and min volume

__Lemma__: $\underline{P} = \{\underline{A}x \le \underline{b}\} \subseteq S(0, R)$

for $R = \sqrt{n} \cdot 2^{\langle A, b \rangle - n^2}$

$\langle A, b \rangle \triangleq$ encoding length of A & b.

__Lemma__: If P is full dim'l, then

$\mathrm{vol}(P) \geq 2^{-(n+1)\langle A \rangle + n^3}$.

Putting it together

- $E_0 = E(A_0, c_0)$, $A_0 = R^2 I$, $\underline{\underline{R}} = \sqrt{n} \cdot 2^{\langle A, b\rangle - n^2}$

  $c_0 = 0$

- $E_k = E(A_k, c_k)$

  If $c_k \in \underline{P}$ STOP

  Else : $\exists\, \hat{a}$ s.t. $\hat{a}^T x \leq \hat{a}^T c_k \quad \forall x \in \underline{P}$

  use $\hat{a}$ to define c_{k+1}, A_{k+1}.

- If $k = \underline{\underline{N}} = 2n\left((2n+1)\langle A\rangle + n\langle b\rangle - n^3\right)$ $\underline{\text{stop}}$

  return "$P = \phi$".

Comments on Computational Requirements

The alg. only needs $\langle A, b \rangle$

to initialize $\left(E_0 = E(R^2 I, 0) \right)$

and to know when we can stop and

declare that $P = \phi$.

circumscribing radius

contained ball

Alg only needs R and r

Discussion

Method presented is called "central cut" ellipsoid.

Other related approaches.

- Shallow cuts
- Deep cuts.